Deep-Tech Frontier Intelligence: AI Hardware

Sample weekly issue, Oct 28, 2026. Five AI-hardware signals across photonics, in-memory compute and advanced packaging, each with a readiness score, patent evidence, global stock exposure and green/red impact.

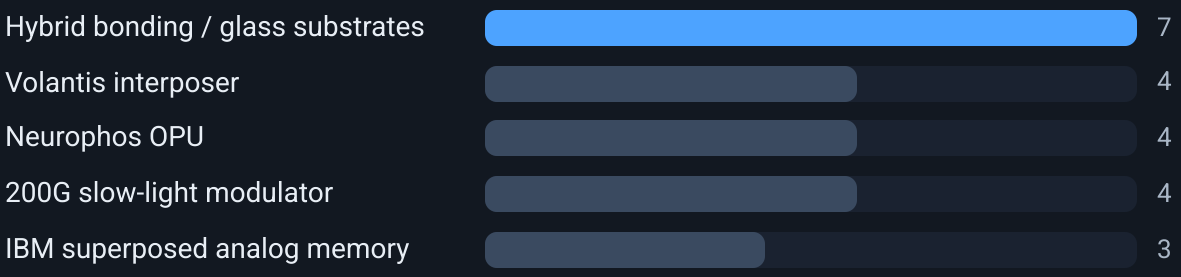
Executive briefing

Bottom line. The money and research this month went into the memory and interconnect wall, not raw compute. Photonic startups now hold real patent estates. Packaging is the layer closest to revenue, and India's listed OSATs are the local way in.

1. Closest to revenue: Applied Materials + Besi hybrid bonding (granted US patents in 2026), plus glass substrates aiming for a production line by end-2027.
2. Strongest IP surprise: Volantis has 26 published US applications and 3 grants for a photonic wafer-scale interposer.
3. IBM's Oct 2026 arXiv paper has a matching patent application, US 2026/0291719 A1, published Sep 24, 2026, by the same authors.
4. India: CG Power (CG Semi) and Kaynes (Kaynes Semicon) are the listed names with OSAT plants in commercial production. Tata Electronics is not listed.
5. Readiness scores are analyst judgment. IP notes are informational, not legal advice. Nothing here is investment advice.

Readiness at a glance

Commercialization readiness, 1–10 (analyst estimate)



Volantis moves up from 3 to 4 compared with the first draft, because its granted patents and published applications are now verified.

1. Neurophos optical processing unit

What	MIT CSAIL's Julia Lab and Neurophos announced a collaboration (Oct 21) to run numerical algorithms on Neurophos' Optical Processing Unit, an optical systolic array for matrix math.
Key claims	Metasurface modulators about 10,000x smaller than Mach-Zehnder devices, with millions of elements in 5x5 mm. The company states a goal of about 50x lower compute resource use. These are company claims.
Why it matters	Optical inference could undercut the energy each GPU operation costs, which is the binding cost in data centers.
Readiness	4/10. The architecture works and is patented, but commercial systems are planned only for 2028.
Patents (verified)	US 11,120,332 B2, granted Sep 14, 2021. US 11,843,182 B2, granted Dec 12, 2023. WO 2021/026152 A1 and EP 4007888 (priority Aug 4, 2019). Assignee: Neurophos.
Who / tickers	Neurophos (private). GlobalFoundries (GFS) is linked only through a summit talk.
Source type	Primary: university announcement plus patent records.

2. Volantis photonic memory interposer

What	Volantis, an Altman-backed startup, is building an accelerator with a photonic interposer that links memory to compute using directly modulated VCSELs and on-chip waveguides.
Key claims	Upwards of 10 TB of memory at up to 240 TB/s per package, according to The Register (Oct 5). Earlier, the company claimed 1 pJ/bit links. These are company claims.
Why it matters	It attacks the memory wall directly and could challenge HBM plus electrical interposers.
Readiness	4/10. The patent estate is real, but the MVP is still 12–18 months out according to reporting.
Patents (verified)	Assignee Volantis Semiconductor Inc.: 26 published applications and 3 grants (latest grant Aug 25, 2026). Examples: US 2026/0119385 A1 (disaggregated memory on a photonic wafer-scale interposer), US 2026/0121756 A1 (quantum-dot VCSEL interposer), US 2026/0219465 A1 (optical wafer-scale switch).
Who / tickers	Volantis (private). No listed partner was named.
Source type	Press interview, cross-checked against patent filings. No peer-reviewed paper was found.

3. IBM superposed analog memory

What	arXiv 2610.26118: many vectors are stored in superposition in one analog memory array and recovered with an in-memory matrix-vector multiply.
Key result	Run on a prototype phase-change memory chip, on recommendation-model and LLM embedding tables. Accuracy was comparable to full-precision software.
Why it matters	It adds a new scaling knob: more effective storage density without more memory cells.
Readiness	3/10. A real-silicon prototype exists, but there is no product roadmap.
Patents (verified)	US 2026/0291719 A1, 'Compressed Memory Using Vector Superposition', filed Mar 24, 2025, published Sep 24, 2026. Applicant IBM. Same inventors as the paper. Background: US 10,678,885 B2 (IBM, in-memory matrix-vector multiply).
Who / tickers	IBM (IBM). Related noise-handling work by Micron (MU) authors: arXiv 2610.20359.
Source type	Primary: preprint plus patent application.

4. 200G slow-light silicon modulator

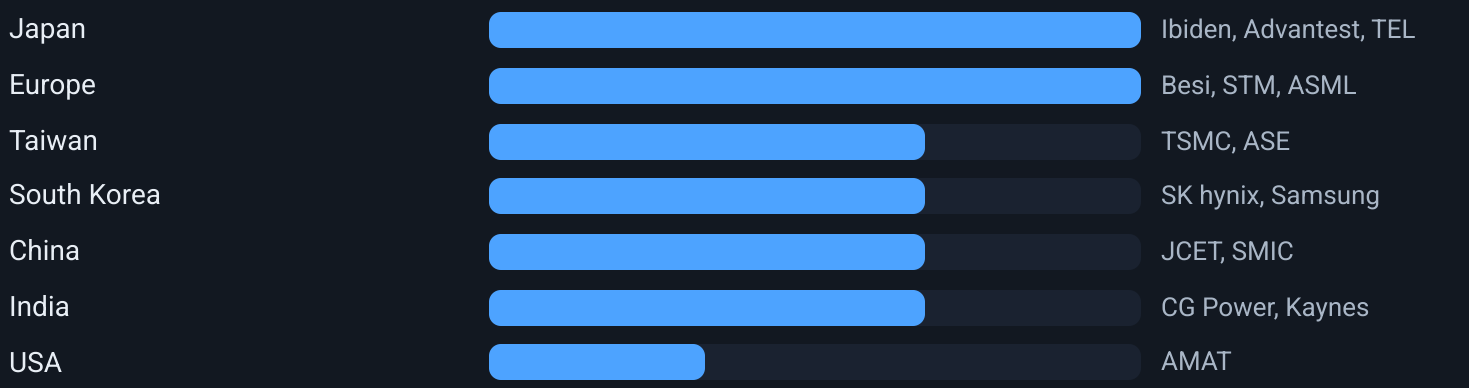
What	Nature Communications (Oct 22): a 320 μm silicon photonic-crystal slow-light modulator with a built-in electro-optic equalizer.
Key result	140 Gb/s OOK at 26 fJ/bit with CMOS-level drive. 180 Gb/s PAM4 without DSP, and 200 Gb/s with an 11-tap equalizer.
Why it matters	It cuts DSP and driver power, a route to dense co-packaged optics for AI clusters.
Readiness	4/10. Lab device only; it is not yet in a co-packaged optics product.
Patents	No patent from this author group was verified in this run. Prior work: Baba group at Yokohama National University, with NTT-affiliated co-authors. Run a family search before diligence.
Who / tickers	The paper names no company commercializing it. Co-packaged optics is in scope for the AMAT–Besi program below.
Source type	Primary: peer-reviewed journal.

5. Hybrid bonding and glass substrates

What	Applied Materials and Besi expanded their partnership on Oct 1. Besi joins Applied's EPIC Center to work on die-to-wafer hybrid bonding, TCB scaling, die-on-panel and co-packaged optics. Their jointly developed Kinex system is already on sale.
Glass cores	Samsung Electro-Mechanics' 100x100 mm glass substrates are in package-level verification, targeting a production line by end-2027. BOE's 24-layer glass substrates are in AI-chip packaging trials.
Why it matters	Large multi-die GPU+HBM packages warp on organic substrates. Glass substrates and hybrid bonding are the way to keep scaling.
Readiness	7/10. A product is shipping (Kinex) and substrates are in customer verification.
Patents (verified)	US 12,550,662 B2, hybrid bonding process flow, filed Nov 18, 2022, granted Feb 10, 2026. US 12,538,855, hybrid bonding process sequencing, filed Dec 9, 2022, gazetted Jan 27, 2026. Both assigned to Applied Materials.
Who / tickers	AMAT (Nasdaq), BESI (Euronext Amsterdam), Samsung Electro-Mechanics (Korea-listed), BOE (China-listed).
Source type	Company press release, product page and patent grants. No academic paper is needed: this is an industry track.

Global chip map

Listed companies with sourced exposure. Tickers from market.vlsi.kr, company IR, or press releases. Not financial advice.



Region	Company	Ticker / exchange	Theme	Sourced exposure
USA	Applied Materials	AMAT, Nasdaq	Hybrid bonding (#5)	Kinex die-to-wafer bonder co-developed with Besi; Oct 1 expansion
Taiwan	TSMC	2330.TW	Glass substrates (#5)	Validating glass-core substrates for CoWoS with Ibiden and Innolux (Chosunbiz, Jun 2026)
Taiwan	ASE	3711.TW	Packaging (#5)	Largest OSAT, 1H26 revenue $7.56B, +33.1%; TGV glass pilot line targeted Q3 2026
South Korea	SK hynix	000660.KS	Hybrid bonding (#5)	Reported first mass-production hybrid-bonding order using AMAT and Besi tools; full-scale use expected around HBM5
South Korea	Samsung Electronics	005930.KS	Hybrid bonding (#5)	Evaluating Besi hybrid bonders; says partial use from HBM4E
Japan	Ibiden	4062, TSE	Glass substrates (#5)	Co-developing glass cores with TSMC; says it holds 70–80% share of AI-server substrates
Japan	Advantest	6857.T	Test for AI chips and photonics (#1, #4)	Record Q2 FY26 (Oct 28); cites strong silicon-photonics and chiplet test demand
Japan	Tokyo Electron	8035.T	Fab tools	CFO sees a record ¥1tn operating profit on AI demand (Nikkei, Oct 7)
Europe	Besi	BESI, Euronext Amsterdam	Hybrid bonding (#5)	Joins Applied's EPIC Center; under evaluation at Samsung
Europe	STMicroelectronics	STM, NYSE	Silicon photonics (#4)	PIC100 in high-volume production for hyperscalers; capacity to quadruple by 2027; TSV version targets co-packaged optics
Europe	ASML	ASML	Lithography	Its earnings were cited by Bernstein as showing AI-driven demand strength
China	JCET	600584, SSE	Glass substrates and CPO (#4, #5)	Pilot lines for glass substrates and panel-level packaging; CPO capability; AI data-center revenue +54% (9M26)

Region	Company	Ticker / exchange	Theme	Sourced exposure
China	SMIC	0981.HK	Foundry and packaging	Expanding advanced packaging for AI chips; raising some advanced-node prices
India	CG Power / Kaynes	NSE: CGPOWER / KAYNES	Packaging (#5)	See India angle below

Not included because no exposure was sourced this pass: Infineon, ASM International, Samsung Electro-Mechanics and BOE (tickers not verified), and Amkor (ranked the #2 OSAT, but ticker not verified). Photonic compute (#1, #2) has no listed pure-play.

Impact map

Green = beneficiary, red = at risk. Scores and cell ratings are our judgment from the cited facts, not financial advice.

Company	Photonic compute	In-memory compute	Glass substrates	Hybrid bonding	Silicon photonics / CPO	Why (fact or judgment)
Applied Materials	–	–	+	++	+	Fact: Kinex bonder, granted bonding patents, CPO in EPIC scope
Besi	–	–	+	++	+	Fact: co-develops Kinex; under evaluation at Samsung
Advantest	+	–	–	–	++	Fact: cites strong silicon-photonics and chiplet test demand
TSMC	–	–	++	+	–	Fact: validating glass for CoWoS. Judgment: bonding helps 3D stacking
ASE	–	–	+	–	–	Fact: glass pilot line; OSAT revenue +33%
STMicroelectronics	-	–	–	–	++	Fact: PIC100 in volume. Judgment: in-package photonics such as Volantis could bypass pluggable optics
JCET	–	–	+	+	+	Fact: glass pilot lines, CPO capability, AI revenue +54%
Ibiden	–	–	+/-	–	–	Fact: co-develops glass with TSMC. Judgment: glass could erode its organic-substrate lead
SK hynix	–	-	–	+	–	Fact: first bonding order. Judgment: photonic memory (Volantis) and analog in-memory compute

Company	Photonic compute	In-memory compute	Glass substrates	Hybrid bonding	Silicon photonics / CPO	Why (fact or judgment)
						could pressure HBM long term
Samsung Electronics	–	-	–	+	–	Same as SK hynix (judgment)
Tokyo Electron	–	–	–	+	–	Fact: record profit outlook on AI. Judgment: benefits from more process steps
SMIC	–	–	–	+	–	Fact: expanding advanced packaging
CG Power / Kaynes	–	–	-	–	–	Fact: wire-bond and substrate OSAT. Judgment: far from glass and hybrid bonding, so they risk being left behind
GPU incumbents (unnamed)	-	-	–	–	–	Judgment: Neurophos targets about 50x efficiency vs GPUs (company claim, systems planned 2028)

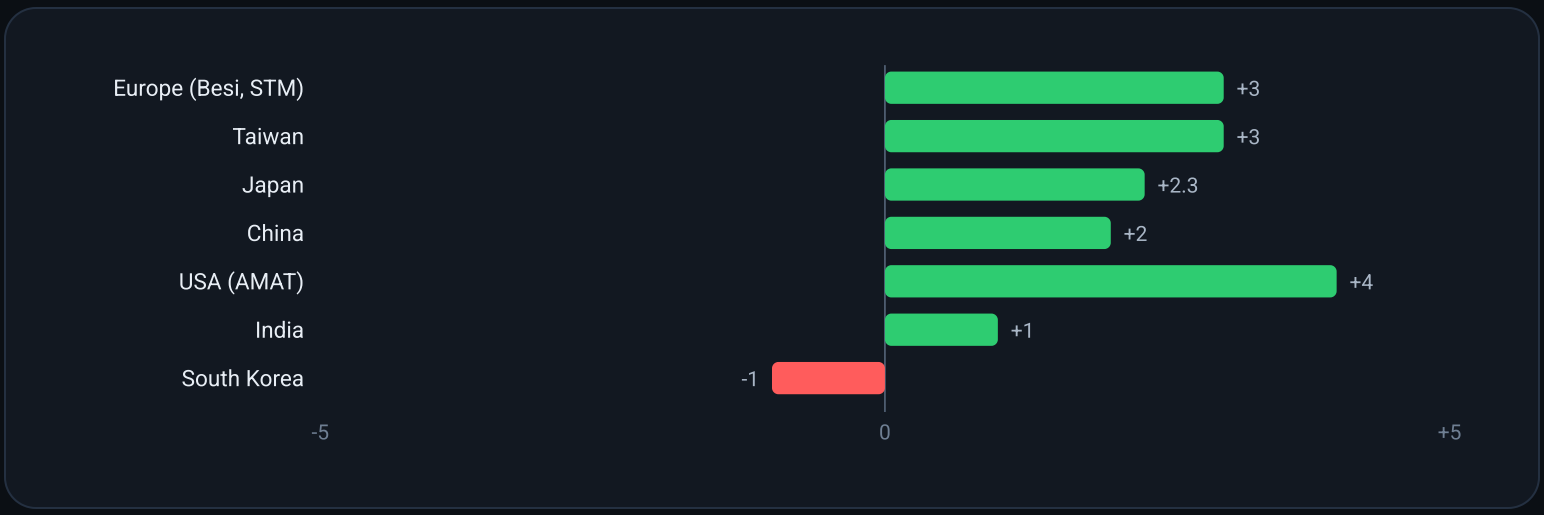
Impact score by company

Our judgment, −5 to +5, as of Oct 28, 2026



Impact score by region

Our judgment: average of the company scores above



Global breakthrough watch

Region	Signal	Date	Why watch
USA	Neurophos and MIT optical computing; Volantis photonic interposer	Oct 21 / Oct 5	Photonic compute leaving the lab
Japan	Advantest record Q2, raised outlook; Tokyo Electron results due Oct 30	Oct 28 / Oct 30	Read on AI test and tool demand into CY2027
Japan	200G slow-light silicon modulator (Nature Communications)	Oct 22	Lower-power optics for co-packaged optics
South Korea	Samsung Electro-Mechanics glass substrates in package verification	Oct 6	Glass line targeted for end-2027
China	JCET Q3: AI revenue +54%; BOE glass in trials	Oct 22 / Oct 16	China building a domestic advanced-packaging stack
Europe	Besi joins Applied's EPIC Center	Oct 1	Hybrid bonding becoming a shared platform
India	CG Semi and Kaynes OSATs producing; Tata Assam under construction	2026	New OSAT supply base

IP signals summary

Informational only, not legal advice or a freedom-to-operate opinion

Pick	Key filings	Assignee	Signal
Neurophos	US 11,120,332; US 11,843,182; EP 4007888	Neurophos	Early priority (2019) on metasurface optical compute; the WO filing has ceased, so check which countries are covered
Volantis	26 US applications, 3 grants	Volantis Semiconductor	Fast, dense filing in 2026 around a wafer-scale photonic interposer; may block followers
IBM	US 2026/0291719 A1; US 10,678,885	IBM	Paper and patent share authors, a sign IBM intends to license or productize
Slow-light modulator	Not verified	—	Open question; check NTT / Yokohama National University families
Hybrid bonding	US 12,550,662; US 12,538,855	Applied Materials	Granted process patents protect the tool flow; Besi is a partner, not a co-assignee, on these

India angle

Only companies whose exposure is in a cited source

Company	Listed as	Exposure	Theme link	Source
CG Power and Industrial Solutions	NSE: CGPOWER	CG Semi OSAT in Sanand, ₹7,584 cr, with Renesas and Stars; production since mid-2026	Packaging (#5)	pib.gov.in PRID 2223049; DSCI report
Kaynes Technology India	NSE: KAYNES	Kaynes Semicon OSAT, ₹3,307 cr, wire-bond and substrate packages; commercial production from Mar 31, 2026	Packaging (#5)	pib.gov.in PRID 2223049; genvest.ai
SPEL Semiconductor	Listed, small cap	Legacy IC assembly and test	Packaging (#5), small scale	genvest.ai
Tata Electronics	Not listed	Assam OSAT (₹27,120 cr) and Dholera fab with PSMC, both under construction	Packaging (#5)	pib.gov.in; DSCI
3D Glass Solutions (Odisha)	Not listed	Approved advanced-packaging and glass-substrate project	Glass substrates (#5)	tensor-advisory.com

> **Read before trading.** Tata Technologies and Tata Elxsi do not give direct ownership of Tata Electronics' plants. Equirus rates CGPOWER and KAYNES 'Reduce' (Business Standard). No listed Indian company was linked to photonic compute or in-memory compute.

How to use this report

Reader	What they get	Example action this week	Tier
VC	Deal flow from patents and papers before funding rounds	Approach Volantis or Neurophos and diligence their patent estates	Teardown $299
Hedge fund / trader	Map from technology to tickers, with readiness timing	Watch AMAT and BESI on hybrid-bonding orders, and CGPOWER and KAYNES on OSAT ramps	Pro $49 or Teardown
Corporate R&D	Prior-art and licensing leads	Pull the Neurophos and IBM patent families before starting optical or in-memory programs	Enterprise
Retail investor	Plain-English trends, with hype flagged	Learn the difference between OSAT names with real plants and 'semiconductor' name-alikes	Free

Subscribe and pricing

Tier	Price	Includes
Free	$0	Weekly top-5 headlines and readiness scores
Pro	$49/mo	Full issue, patent tables, US and India ticker maps
Teardown	$299/mo	Pro plus a monthly 360° technical teardown and deck
Enterprise	Custom, from $2k/mo	Custom watchlists, patent alerts, analyst calls

60-second video script

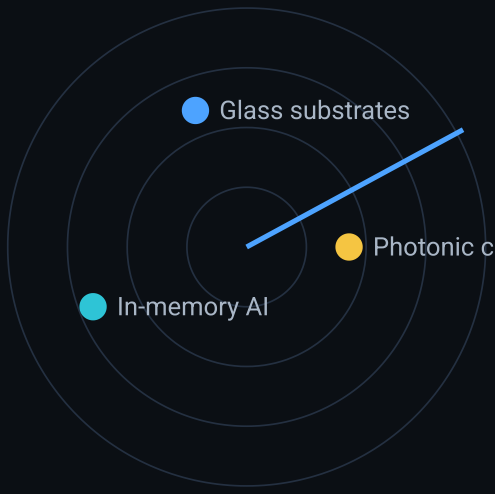
1. **0–8s hook:** AI's biggest bottleneck isn't compute anymore. It's moving data. Here are five moves from this month.
2. **8–22s photonics:** Neurophos puts millions of optical elements in 5 millimeters, backed by patents dating to 2019. Volantis has 26 patent applications for moving memory over light.
3. **22–32s memory:** IBM stored multiple AI vectors in superposition on one phase-change chip, and filed a patent on it.
4. **32–42s interconnect:** A new silicon modulator reaches 200 gigabits per lane at 26 femtojoules per bit.
5. **42–55s packaging and India:** Applied Materials and Besi are expanding hybrid bonding. In India, CG Semi and Kaynes Semicon are already producing chips.
6. **55–60s close:** Scores, patents and tickers are in this week's Frontier Intelligence brief. Not investment advice.

Sources

1. Julia Lab and Neurophos collaboration
2. US11843182B2, Neurophos
3. WO2021026152A1, Neurophos
4. Patents assigned to Neurophos LLC (US 11,120,332)
5. Volantis photonic interposer plan (The Register)
6. Volantis Semiconductor assignee profile
7. Volantis filings by Tapabrata Ghosh
8. Superposed analog memory for vector storage (arXiv 2610.26118)
9. IBM US 2026/0291719 A1 (Patentlyze)
10. IBM US 10,678,885 B2
11. Weight splitting on noisy in-memory accelerators (Micron, arXiv 2610.20359)
12. 200G/λ slow-light silicon modulator (Nature Communications)
13. Applied Materials and Besi expand partnership
14. US20240170443A1 / US 12,550,662
15. US 12,538,855 (USPTO Official Gazette)
16. Kinex hybrid bonding system
17. Samsung Electro-Mechanics glass substrates (MoneyToday)
18. BOE 24-layer glass substrates (DIGITIMES)
19. Approved semiconductor projects (PIB India)
20. OSAT stocks in India 2026 (Genvest)
21. CG Power and Kaynes listed exposure (Business Standard)
22. TSMC glass substrate validation with Ibiden (Chosunbiz)
23. Ibiden 1Q FY2027 earnings call (Envalith)
24. Ibiden interim Q&A
25. Samsung and SK hynix hybrid bonder timing (Chosunbiz)
26. SK hynix hybrid bonding at Hot Chips 2026 (Tom's Hardware)
27. Advantest Q2 FY2026 results
28. Advantest earnings call (Investing.com)
29. Tokyo Electron record profit outlook (Nikkei Asia)
30. Bernstein on Japanese chip stocks (Investing.com)
31. ST PIC100 silicon photonics in high-volume production
32. JCET Q3 YTD 2026 results
33. China advanced packaging and OSAT ranking (eefocus)
34. SMIC returns to advanced packaging (DIGITIMES)
35. Glass core substrate sector research (ASE pilot line)
36. Chip company ticker list (market.vlsi.kr)
37. State of India's Semiconductor Industry 2026 (DSCI)
38. India semiconductor project map (Tensor Advisory)

IP notes are informational, not legal advice. Readiness scores are analyst judgment. Performance figures are company claims as reported. Patent statuses come from public databases and may lag.

Not financial advice. Scores and impact ratings are our judgment. IP notes are informational, not legal advice. Company performance figures are company claims as reported. © 2026 FrontierRadar.



FrontierRadar

See the breakthrough before the market does.

A weather radar for technology: papers + patents in, scored signals

A weather radar spots a storm before it hits. FrontierRadar spots technology storms (a paper, a patent wave, a factory ramp) before they show up in earnings, and gives each one a readiness score.

Pipeline



Value path 1: Investing anywhere, plus an India angle

1. **Spot the theme:** Sample report: glass substrates and chip packaging scored 7/10, the highest.
2. **Map it globally:** USA: AMAT. Taiwan: TSMC 2330.TW, ASE 3711.TW. Korea: SK hynix 000660.KS, Samsung 005930.KS. Japan: Ibiden 4062, Advantest 6857.T. Europe: Besi, STM. China: JCET 600584. Readers buy through their own local broker.
3. **India leg:** CG Power (CG Semi OSAT) and Kaynes (Kaynes Semicon OSAT), both in commercial production in 2026.
4. **Set triggers:** Watch for order announcements, OSAT capacity ramps, and the Samsung Electro-Mechanics glass line (target end-2027).
5. **Act through your broker:** Research and place orders yourself through your own broker, with your own sizing and stop rules.

> **Example, not a recommendation.** This shows the workflow only. Equirus rates CG Power and Kaynes 'Reduce' (Business Standard). This is not financial advice.

Value path 2: Customer revenue

Prices are assumptions to test

Free

$0

Weekly top 5 and scores

Top of the funnel

Pro

$49/mo

Full issue, patents, US and India tickers

Traders and retail

Teardown

$299/mo

Plus monthly 360° teardown and deck

VCs and funds

Enterprise

$2k+/mo

Custom watchlists, alerts, analyst calls

Corporate R&D

Revenue calculator

Example math, not a forecast. Enterprise counted at $2,000/mo

Free readers

Pro subscribers

Teardown subscribers

Enterprise clients

Monthly revenue

$14,880

Annual revenue

$178,560

Free-to-paid ratio

12.0%

Who pays: pain to gain

Persona	Pain	Gain	Tier
Retail investor (any country)	Hype and 'semiconductor' name-alike stocks	Plain-English scores, verified exposure, local tickers	Free / Pro
Global trader	Hears about themes after the move	A 7-region ticker map: US, Taiwan, Korea, Japan, Europe, China, India	Pro
India trader	Few direct chip plays	Verified CGPOWER and KAYNES exposure, with name-alikes flagged	Pro
VC	Finds startups at the same time as everyone else	Startups surfaced from patents before the funding round	Teardown
Fund	Analysts can't read every paper	Weekly scored summary with citations	Teardown / Enterprise
Corporate R&D	Prior-art blind spots	Patent families and licensing leads	Enterprise

Global coverage map

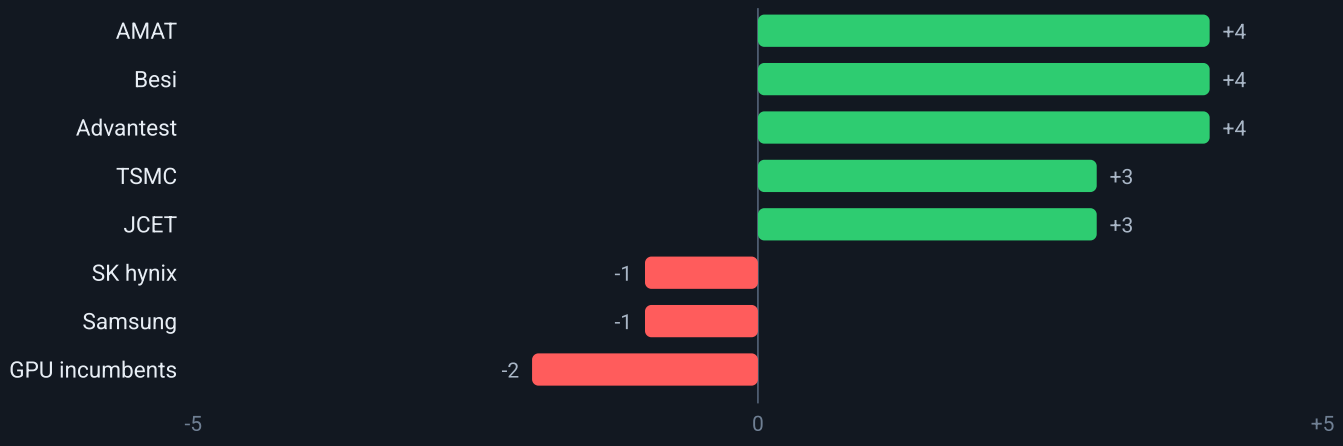
Names per region from the sample report's global chip map

Japan	Ibiden, Advantest, TEL
Europe	Besi, STM, ASML
Taiwan	TSMC, ASE
South Korea	SK hynix, Samsung
China	JCET, SMIC
India	CG Power, Kaynes
USA	AMAT

Region	Watch this week
USA	Neurophos with MIT; Volantis interposer
Japan	Advantest record Q2 (Oct 28); TEL results Oct 30
Korea	Glass substrates in package verification
China	JCET AI revenue +54%
Europe	Besi joins Applied's EPIC Center
India	CG Semi and Kaynes OSATs producing

Market impact snapshot

Our judgment of company impact, from the sample report's Impact map. Green = beneficiary, red = at risk.



Competitive landscape

Player	What it does (sourced)	Pricing (sourced)	Gap we fill
CB Insights	Market intelligence on private companies, funding, M&A and emerging tech	Seat-based terminal plus consumption API; pricing not public	Science-first: we start from papers and patents, not deals
AlphaSense	AI search over premium business documents for finance and corporate teams	Quote-based; third-party data puts entry around $12k/yr	Readiness scoring at a retail-to-mid price
Elicit	AI literature review across academic papers	Plus $12/mo, Pro $49/mo (third-party listing)	Links findings to patents, tickers and timing

AlphaSignal was not verified in this pass, so it is left out.

Our moat

Patent-paper fusion

Example: IBM's arXiv paper matched to its US 2026/0291719 A1 filing

Readiness scoring

A 1–10 score with reasoning, tracked week over week

Global + India angle

7-region ticker map with verified exposure, flagging name-alikes

Multi-format

One analysis becomes a brief, a deck, a video and an X post

90-day launch roadmap

Phase	Days	Deliverables	Exit test
Prove	1–30	4 weekly issues on AI hardware, free newsletter, X presence	200 free readers
Monetize	31–60	Launch Pro $49, first teardown, patent alerts	10 paying subscribers
Scale	61–90	Add a second niche (quantum or bio), 3 enterprise pilots, sponsor deck	First enterprise pilot

Risk vs opportunity

Our judgment of size, −5 to +5. Green = opportunity, red = risk. Not a forecast.



Risks and mitigations

Risk	Mitigation
Hallucinated citations	Every claim is linked, and a human spot-checks each issue
Seen as legal or financial advice	'Not advice' labels, IP signals only, no buy or sell calls
Token and compute cost	Weekly cadence, one niche first
Incumbents copy the format	Speed plus India depth plus a patent-paper archive
Slow paid conversion	Free tier first, sponsorship revenue as a bridge

KPIs to track

- Free subscribers and weekly open rate (target 40%+ opens)
- Free-to-paid conversion (target 2–5%)
- Monthly recurring revenue and churn
- Citation error rate (target 0 per issue)
- Signal hit rate: scored themes that later hit news or earnings
- ✔ Sample report #1 published as a Page (as of Oct 28, 2026)

Get the weekly radar

Free weekly top 5 with readiness scores.

Placeholder: no data is collected yet.

Sources

- Sample weekly report: AI Hardware, Oct 28, 2026 (all company and patent sources)
- CB Insights pricing
- AlphaSense pricing
- AI market research tools (Elicit, AlphaSense pricing)

Not financial advice. Revenue figures are illustrative math. Prices are assumptions. Competitor details come from the linked pages as of Oct 28, 2026.

Not financial advice. Scores and impact ratings are our judgment. IP notes are informational, not legal advice. Company performance figures are company claims as reported. © 2026 FrontierRadar.